CLOUD SECURITY: ENCRYPTION STORAGE AT REST

by Matteo Salis

ENCRYPTION STORAGE

```
❯ openssl enc -aes-256-cbc -pbkdf2 -k "test" -in prova.txt -out prova.enc -v
bufsize=8192
bytes read   :       40
bytes written:       64
❯ openssl enc -aes-256-cbc -pbkdf2 -k "test" -in prova.txt -out prova.enc -p
salt=D4EF135E9FE0BB86
key=CDC58341C76AD559E45CA7498A14E1E99DC3783BE828F4446D884F84E11B8C29
iv =C83FCE21916CE34C5F6E78ADDEF024C6
❯ xxd prova.enc
00000000: 5361 6c74 6564 5f5f d4ef 135e 9fe0 bb86  Salted__...^....
00000010: 9c31 1c75 26ea 361c b785 8ada 03f8 77ad  .1.u&.6.......w.
00000020: c84c a2c3 78fa 8b0c ee93 2d75 3b7e 657e  .L..x.....-u;~e~
00000030: 76fc e858 e62d cdb0 4e21 0427 efd2 2879  v..X.-..N!.'..(y
```

- LUKS (Linux Unified Key Setup)
- CryFS – Crittografia pensata per Cloud Storage

AES (ADVANCED ENCRYPTION STANDARD)

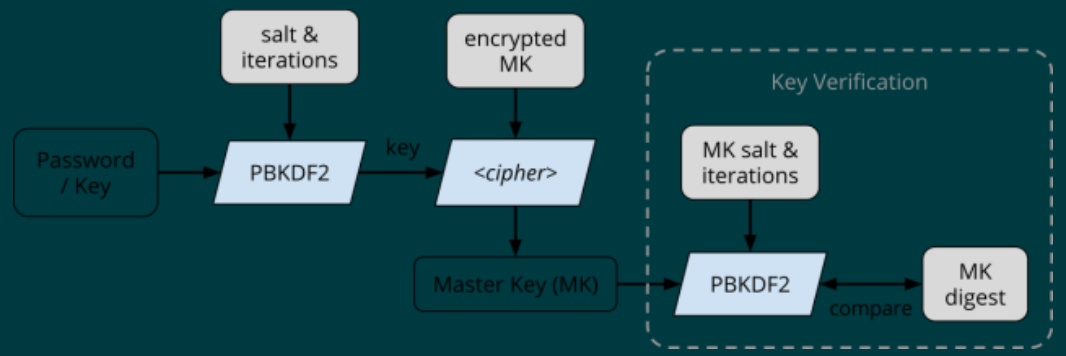
Introdotto nel 1997, si tratta di un algoritmo simmetrico a blocchi progettato per gestire chiavi di lunghezza variabile, precisamente di 128, 192 e 256 bit, garantendo così un elevato livello di sicurezza crittografica.

128 = 16 byte

192 = 24 byte

256 = 32 byte

LUKS (LINUX UNIFIED KEY SETUP)



Luks è uno standard generico per la **gestione della crittografia dei dischi,** progettato specialmente con il sistema di crittografia "dm-crypt" su Linux.

PBKDF1 e PBKDF2 sono funzioni di derivazione chiavi per rendere più costosi gli attacchi brute-force su password — PBKDF1 aveva un limite di 160 bit ed è superata — PBKDF2 la sostituisce ed è specificata in PKCS #5 v2.0 (RFC 2898) e raccomandata in RFC 8018 (PKCS #5 v2.1, 2017)

```
❯ sudo cryptsetup luksFormat /dev/sda
WARNING: Device /dev/sda already contains a 'ext4' superblock signature.

WARNING!
========
This will overwrite data on /dev/sda irrevocably.

Are you sure? (Type 'yes' in capital letters): YES
Enter passphrase for /dev/sda:
Verify passphrase:
❯ sudo cryptsetup open /dev/sda cryptdata
Enter passphrase for /dev/sda:
❯ sudo cryptsetup status cryptdata
/dev/mapper/cryptdata is active.
  type:    LUKS2
  cipher:  aes-xts-plain64
  keysize: 512 [bits]
  key location: keyring
  device:  /dev/sda
  sector size:  512 [bytes]
  offset:  32768 [512-byte units] (16777216 [bytes])
  size:    15941632 [512-byte units] (8162115584 [bytes])
  mode:    read/write
❯ lsblk -f
NAME        FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda         crypto 2           8a72bf5a-b378-41c6-80c5-6874f7b16c31
└─cryptdata
nvme0n1
├─nvme0n1p1 vfat   FAT32       D221-87A9                                 1G     3% /boot
├─nvme0n1p2 swap   1           fc2b1390-a516-4c4b-85e2-c8597f4124de                [SWAP]
└─nvme0n1p3 ext4   1.0         15be61b9-a9b7-4b76-b7f9-5974bae58cbd   21.5G    90% /
```

HEADER LUKS

- ID: magic, ver., UUID
- Crypto: cipher/mode/hash + offset
- Chiavi: PBKDF2 + 8 key slot

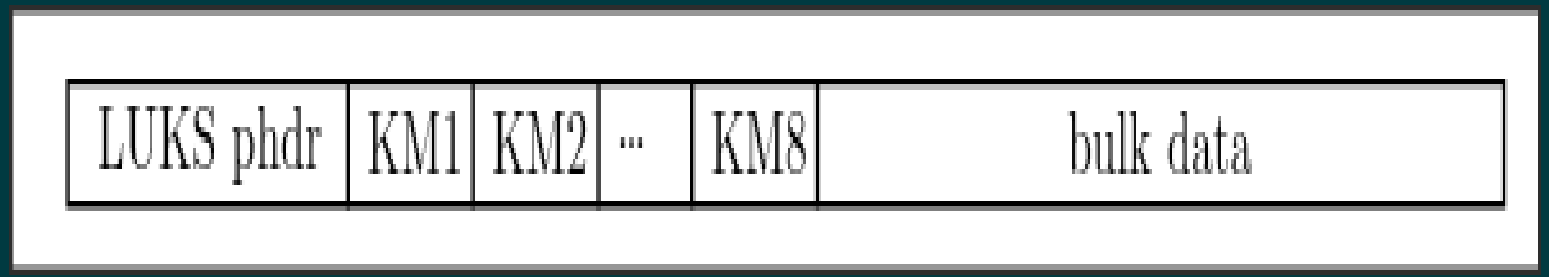
```
59 3b 7d 16 28 e0 21 7c bc 05 e5 65 cd d3 78 6e
❯ sudo cryptsetup luksAddKey /dev/sda
Enter any existing passphrase:
Enter new passphrase for key slot:
Verify passphrase:
❯ sudo cryptsetup luksDump /dev/sda
LUKS header information
Version:        2
Epoch:          4
Metadata area:  16384 [bytes]
Keyslots area:  16744448 [bytes]
UUID:           8a72bf5a-b378-41c6-80c5-6874f7b16c31
Label:          (no label)
Subsystem:      (no subsystem)
Flags:          (no flags)

Data segments:
  0: crypt
        offset: 16777216 [bytes]
        length: (whole device)
        cipher: aes-xts-plain64
        sector: 512 [bytes]

Keyslots:
  0: luks2
        Key:        512 bits
        Priority:   normal
        Cipher:     aes-xts-plain64
        Cipher key: 512 bits
        PBKDF:      argon2id
        Time cost:  8
        Memory:     1048576
        Threads:    4
        Salt:       2b 1f d4 83 07 98 24 cf 3d 0f 16 d9 ad f4 f3 9b
                    3e 2b 3e 28 96 33 56 ba d8 35 84 24 af 51 0b b2
        AF stripes: 4000
        AF hash:    sha256
        Area offset:32768 [bytes]
        Area length:258048 [bytes]
        Digest ID:  0
  1: luks2
        Key:        512 bits
        Priority:   normal
        Cipher:     aes-xts-plain64
        Cipher key: 512 bits
        PBKDF:      argon2id
        Time cost:  8
        Memory:     1048576
        Threads:    4
        Salt:       43 6c 43 3d ff 64 33 a1 ec 3b 32 fc 02 ba c8 90
                    f8 25 1f 51 0d 7d fb 8e ee e3 9b 57 5a 0a 43 d0
        AF stripes: 4000
        AF hash:    sha256
        Area offset:290816 [bytes]
        Area length:258048 [bytes]
        Digest ID:  0
Tokens:
Digests:
  0: pbkdf2
        Hash:       sha256
        Iterations: 490906
        Salt:       d3 34 7a e2 1b 21 0f c4 29 94 28 89 f3 6b 69 d6
                    3d 00 79 a2 64 78 21 fe 9e 11 c9 ef 75 32 04 4c
        Digest:     94 6a 3a a8 f8 38 44 a9 1b 87 2f 88 4d a8 dc 02
                    59 3b 7d 16 28 e0 21 7c bc 05 e5 65 cd d3 78 6e
```

Luks 1	Luks 2
Struttura binaria rigida	Struttura flessibile con JSON
Dimesione 592 byte	Dimesione scalabile
Limite 8 slots da 42 byte	Slot flessibili
Nessuna ridondanza	Backup multipli di metadati
Supporta solo PBKDF2	Supporta algoritmi piu moderni

LUKS1 layout

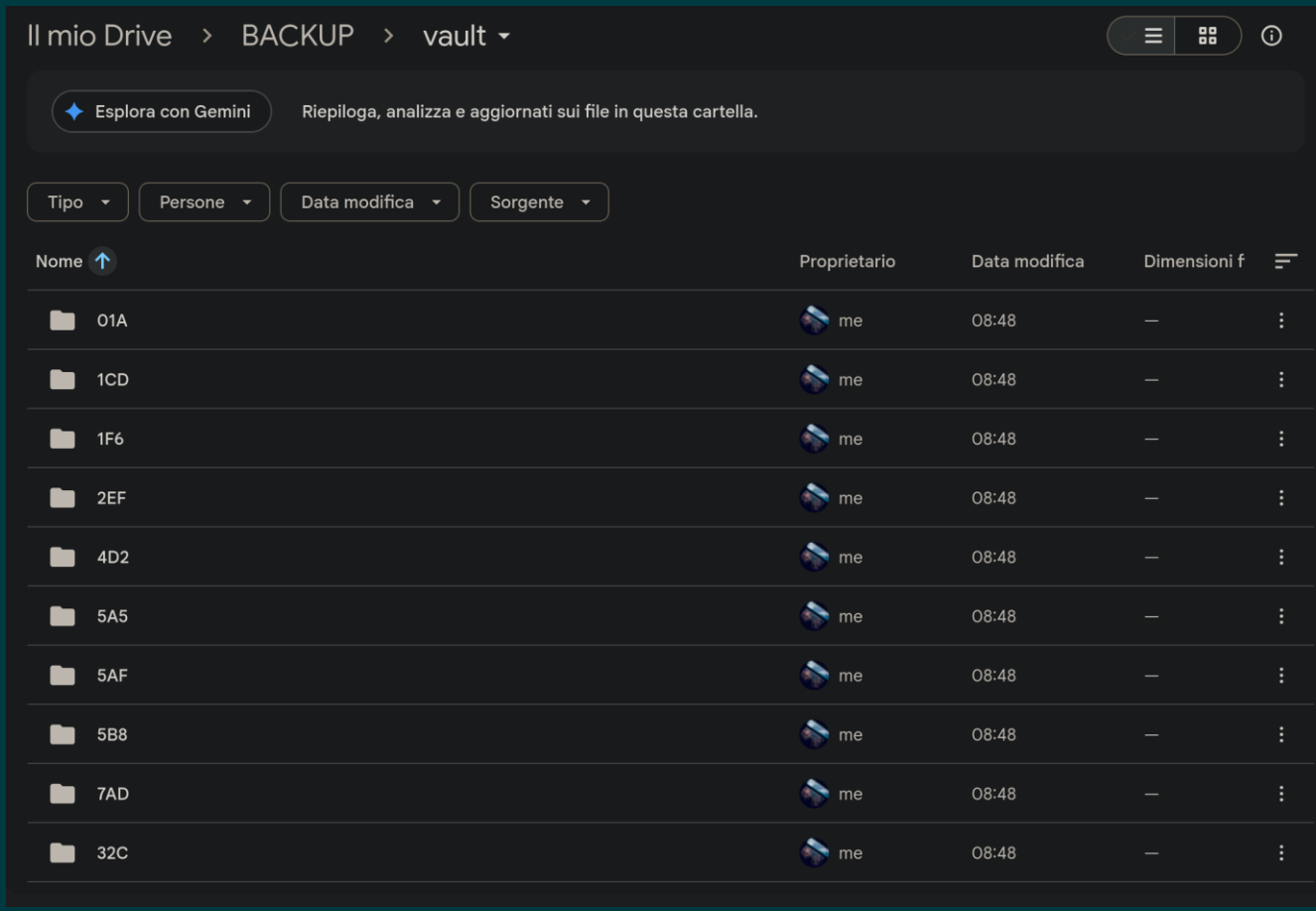


LUKS2 layout

CRYFS: CIFRATURA CLIENT-SIDE CLOUD

CryFS è un filesystem crittografico pensato per il cloud. Cifra non solo il contenuto dei file, ma anche dimensione, metadati e struttura delle cartelle.

- Divide ogni file in blocchi da 16KB
- Ogni blocco viene cifrato separatamente con **xchacha20-poly1305**
- Ogni blocco salvato con un **nome casuale** (es. **01A**, **4D2**, **5B8**)
- Risultato: nel cloud non si vede né il contenuto, né i nomi, né la struttura



PRATICA CRYFS

```
❯ cryfs ~/GDrive/vault ~/GDrive/mnt
CryFS Version 1.0.3+0.g5d0cc3678.dirty
WARNING! This is a development version based on git commit 5d0cc3678. Ple
ase do not use in production!

Could not find base directory. Do you want to create it?
Your choice [y/n]: y
Could not find mount directory. Do you want to create it?
Your choice [y/n]: y
Use default settings?
Your choice [y/n]: y

Generating secure encryption key. This can take some time ... done
Password:
Confirm Password:
Deriving encryption key (this can take some time) ... done

------------------------------------------------------
Filesystem configuration:
------------------------------------------------------
- Filesystem format version: 0.10
- Created with: CryFS 1.0.3+0.g5d0cc3678.dirty
- Last opened with: CryFS 1.0.3+0.g5d0cc3678.dirty
- Cipher: xchacha20-poly1305
- Blocksize: 16384 bytes
- Filesystem Id: 376B23B5F064D8A30A693B80A49090DC
------------------------------------------------------

Mounting filesystem. To unmount, call:
$ cryfs-unmount "/home/matte/GDrive/mnt"
```

```
❯ ls GDrive/vault/
01A/  2EF/  38C/  5A5/  68C/  89A/  BEB/  CC2/          D4F/  E23/
1CD/  312/  420/  5AF/  75A/  A50/  C1A/  CD0/          D6D/  F3D/
1F6/  32C/  4D2/  5B8/  7AD/  BC3/  CA9/  cryfs.config  DA3/
```

SETUP BENCHMARK

- OS: Arch
- Tool: **dd**
- Parametri: blocchi da 1MB, 1024 (= 1GB totale)

```
❯ dd if=/dev/zero of=testfile bs=1M count=1024 oflag=direct
1024+0 records in
1024+0 records out
1073741824 bytes (1.1 GB, 1.0 GiB) copied, 0.518076 s, 2.1 GB/s
❯ dd if=testfile of=/dev/null bs=1M iflag=direct
1024+0 records in
1024+0 records out
1073741824 bytes (1.1 GB, 1.0 GiB) copied, 0.992936 s, 1.1 GB/s
```

```
❯ sudo dd if=/dev/zero of=/mnt/crypto/testfile bs=1M count=1024 oflag=direct
1024+0 records in
1024+0 records out
1073741824 bytes (1.1 GB, 1.0 GiB) copied, 0.512499 s, 2.1 GB/s
❯ sudo dd if=/mnt/crypto/testfile of=/dev/null bs=1M iflag=direct
1024+0 records in
1024+0 records out
1073741824 bytes (1.1 GB, 1.0 GiB) copied, 0.208244 s, 5.2 GB/s
```

BENCHMARK RISULTATI

	No Cifratura	LUKS
Scrittura	2.1 GB/s	2.1 GB/s
Lettura	1.1 GB/s	5.2 GB/s
Overhead	-	0%

	LUKS	CryFS
Livello di Cifratura	A livello di Blocco (Block-level)	A livello di File (Client-side / Cloud)
Caso d'uso	Dischi fisici, partizioni di sistema, SSD interni, chiavette USB	Cloud Storage (Google Drive, Dropbox, ecc.)
Gestione dei Dati	Cifra l'intero dispositivo/partizione in un blocco unico continuo.	Divide ogni file in blocchi da 16KB cifrati separatamente (con xchacha20-poly1305).
Protezione Privacy	Cifra i dati fisici. Il cloud vedrebbe un unico gigantesco file illeggibile di dimensioni fisse.	Nel cloud non si vede né il contenuto, né i nomi dei file, né la struttura delle cartelle.
Prestazioni (I/O)	Alte	Inferiori

LUKS VS CRYFS